Testiddle

PRACTICE TEST

01

TT-SAT-01



SAT-STYLE PRACTICE SERIES

SAT
Practice Test

Digital format, printed. Sit it in one go, on the clock, then mark it and learn from every miss.

4	120	164
MODULES	QUESTIONS	MINUTES

Before you start

- [ ] A sharp pencil, an eraser, and a watch or timer
- [ ] A calculator for the two Math modules
- [ ] A quiet table and about 184 minutes with no interruptions
- [ ] Circle your answers in this booklet, or use the answer sheet at the back

MODULE	QUESTIONS	TIME
Reading and Writing, Module 1	33	39 min
Reading and Writing, Module 2 (harder)	33	39 min
Math, Module 1 (calculator permitted)	27	43 min
Math, Module 2 (harder)	27	43 min

Every question in this test is original and written by Testiddle. It follows the common patterns of the real exam but is not affiliated with, or endorsed by, any exam provider.

Test begins on the next page.

Start your timer now.

1 Reading and Writing

33 QUESTIONS 39 MINUTES

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer. Circle your answer in the booklet.

PACE YOURSELF

You have **39 minutes** for **33 questions**, which is about **71 seconds** a question. Answer what you can quickly, circle the ones that need more time, and come back to them before time is called. There is no penalty for a wrong answer, so never leave a question blank.

WHAT YOU WILL SEE IN THIS MODULE

- Words in context
- Text structure and purpose
- Cross-text connections
- Central ideas and details
- Inferences
- Command of evidence: textual
- Command of evidence: quantitative
- Boundaries
- Form, structure, and sense
- Transitions
- Rhetorical synthesis

Questions 1 to 4 of 33

1 WORDS IN CONTEXT

Marine biologist Teodora Ilves notes that the deep-sea anglerfish uses a glowing lure to _____ prey in the near darkness: the light draws smaller fish close enough to be caught.

Which choice completes the text with the most logical and precise word or phrase?

(A) repel
(B) attract
(C) imitate
(D) conceal

2 WORDS IN CONTEXT

Although critics praised the novel's ambitious structure, many readers found its constantly shifting timelines more _____ than rewarding, and abandoned the book after the first hundred pages.

Which choice completes the text with the most logical and precise word or phrase?

(A) disorienting
(B) engaging
(C) predictable
(D) concise

3 WORDS IN CONTEXT

Urban planner Marisol Quiñones argues that a city's parks are not merely decorative but _____ : by giving residents a place to gather, they strengthen the social ties that make neighbourhoods resilient.

Which choice completes the text with the most logical and precise word or phrase?

(A) fragile
(B) obscure
(C) functional
(D) excessive

4 WORDS IN CONTEXT

When the museum unveiled a restored 1920s mural, the curators were careful to _____ the artist's original palette, even though modern pigments would have been cheaper and easier to apply.

Which choice completes the text with the most logical and precise word or phrase?

(A) abandon
(B) exaggerate
(C) conceal
(D) preserve

Questions 5 to 6 of 33

5 WORDS IN CONTEXT

The mayor's response to the report was _____ : she neither accepted its conclusions nor dismissed them, saying only that the city would study the findings further.

Which choice completes the text with the most logical and precise word or phrase?

(A) noncommittal

(B) enthusiastic

(C) hostile

(D) apologetic

6 TEXT STRUCTURE AND PURPOSE

The following text is from an original short story. Odile has taken over her father's job as a lamplighter.

> Every evening at dusk, Odile walked the length of Harbour Row with the long brass pole her father had carried before her, touching its small flame to each lamp in turn. The work had been his for thirty years, and the lamps seemed to remember his rhythm, flaring up half a second before she reached them. Odile told herself she had taken the job only until he recovered. But as the last lamp caught and the street glowed amber behind her, she felt something she could not have admitted aloud: that she would be sorry on the day he asked for the pole back.

Which choice best states the main purpose of the text?

(A) To explain how street lamps were lit in the past

(B) To suggest that Odile resents her father for being ill

(C) To show that Odile has come to value a task she believes is only temporary

(D) To describe the changes that have taken place on Harbour Row

7 TEXT STRUCTURE AND PURPOSE

The following text is from an original poem.

> *The morning pours its pale light on the sill,*
> *and I, who meant to sleep, lie listening*
> *to kettles, birds, a neighbour's radio,*
> *the ordinary orchestra of things.*
> *I thought that quiet was the thing I wanted.*
> *But here, inside the clatter, I am held,*
> *as though the whole bright street had come to say*
> *that I am not alone, and never was.*

Which choice best describes the overall structure of the text?

(A) The speaker recalls a past event, then compares it with a present one

(B) The speaker complains about a disturbance, then decides to ignore it

(C) The speaker asks a series of questions, then answers each in turn

(D) The speaker lists a set of everyday sounds, then reveals an unexpected feeling they produce

8 TEXT STRUCTURE AND PURPOSE

The following text is adapted from an original science article about bridge engineering.

> Engineers designing long-span bridges must account for more than the weight of traffic. Wind, too, can destroy a bridge: in 1940, a bridge in Washington State twisted itself apart in a moderate gale, to the astonishment of its designers. Since then, builders of long bridges routinely test scale models in wind tunnels before construction begins.

Which choice best describes the function of the underlined sentence in the text as a whole?

(A) It provides a historical example that explains why a modern practice became standard.

(B) It questions whether wind is a serious threat to bridges.

(C) It introduces a disagreement among engineers about wind tunnels.

(D) It describes the usual design of a long-span bridge.

9

CROSS-TEXT CONNECTIONS

TEXT 1

Many economists argue that remote work raises productivity because employees avoid long commutes and constant interruptions. In one study of a call centre, staff who worked from home completed 13 percent more calls than colleagues in the office.

TEXT 2

A researcher who studied software teams found that remote work did not raise output evenly. Teams whose tasks depended on frequent, unplanned conversations produced less when their members were dispersed, even though individuals reported feeling less distracted.

Based on the texts, how would the researcher in Text 2 most likely respond to the economists' claim in Text 1?

(A) By agreeing that remote work raises productivity in every setting

(B) By arguing that the claim overlooks kinds of work that depend on spontaneous interaction

(C) By denying that employees feel less distracted at home

(D) By recommending that call centres close their offices

10

CENTRAL IDEAS AND DETAILS

Researchers once assumed that the vivid patterns on certain tropical frogs served mainly to attract mates. Field observations, however, show that snakes and birds quickly learn to avoid the brightest individuals, suggesting that the colours function chiefly as a warning. Potential mates, meanwhile, appear to pay little attention to them.

Which choice best states the main idea of the text?

(A) Bright colours on some tropical frogs appear to warn off predators rather than to attract mates.

(B) Predators of tropical frogs have become scarce in recent decades.

(C) Female frogs strongly prefer males with the brightest patterns.

(D) Field observations of tropical frogs are too difficult to be reliable.

11 CENTRAL IDEAS AND DETAILS

The following text is from an original short story. Marcus is a new student.

> Marcus had decided on the first day that the cafeteria was a place to be endured, not enjoyed. He took the table by the window, opened his lunch, and watched the rain stitch lines down the glass. When a girl he barely knew set her tray down opposite him and said nothing at all, he was too surprised to object. They ate in a silence that, to his astonishment, did not feel like loneliness.

Which choice best states the main idea of the text?

(A) Marcus resents a classmate for disturbing his lunch.

(B) Marcus has decided to stop eating in the cafeteria.

(C) Marcus unexpectedly finds comfort in a shared silence that he assumed would feel lonely.

(D) A classmate invites Marcus to join her group of friends.

12 INFERENCES

Chef Amara Okoye has spent three years documenting regional recipes that exist only as spoken instructions, passed from one cook to the next. Because those instructions rely on gestures and taste rather than measured quantities, Okoye's written versions, therefore, _____

Which choice most logically completes the text?

(A) can only approximate the dishes as they were originally made.

(B) are more accurate than the spoken instructions were.

(C) prove that the dishes are not worth preserving.

(D) show that measured quantities were once widely used.

13 INFERENCES

Paleontologists long believed that the long necks of sauropod dinosaurs evolved so the animals could reach high treetop foliage. Yet recent analysis of fossilized neck bones suggests the necks were usually held low and swung from side to side, letting the animals graze a wide area without moving their heavy bodies. If this analysis is accurate, then _____

Which choice most logically completes the text?

(A) sauropods probably did not eat plants.

(B) sauropods were unable to move their heads.

(C) fossilized neck bones are rarely found.

(D) the long necks may have served a purpose other than reaching treetops.

14 COMMAND OF EVIDENCE: TEXTUAL

Marine ecologist Lucía Ferrante hypothesizes that sea otters indirectly protect kelp forests: by eating sea urchins, which graze on kelp, otters keep urchin numbers low enough for the kelp to thrive.

Which finding, if true, would most directly support Ferrante's hypothesis?

A) Sea otters in some areas also eat clams and crabs.

B) Kelp forests grow more slowly in colder water.

C) Urchins in kelp-poor areas tend to be smaller than urchins in kelp-rich areas.

D) In coastal areas where otters were removed, urchin populations grew and kelp forests shrank.

15 COMMAND OF EVIDENCE: TEXTUAL

A literature professor claims that characters in short stories often reveal their true feelings through small, indirect actions rather than through what they say.

Which quotation from a short story would best illustrate the professor's claim?

A) "Dara had been angry for years, and she told everyone so at every opportunity."

B) "The meeting began at noon, and Dara arrived early to set up the chairs."

C) "'I'm perfectly fine,' Dara said, and went on stacking the chairs, though she set each one down a little harder than the last."

D) "Dara loved the town and often said it was the best place in the world to grow up."

16 COMMAND OF EVIDENCE: TEXTUAL

A city official argues that a new subway line caused a rise in restaurant openings along its route, noting that the number of new restaurants near stations doubled within two years of the line opening.

Which finding, if true, would most seriously weaken the official's argument?

A) Many of the new restaurants near stations serve food from other countries.

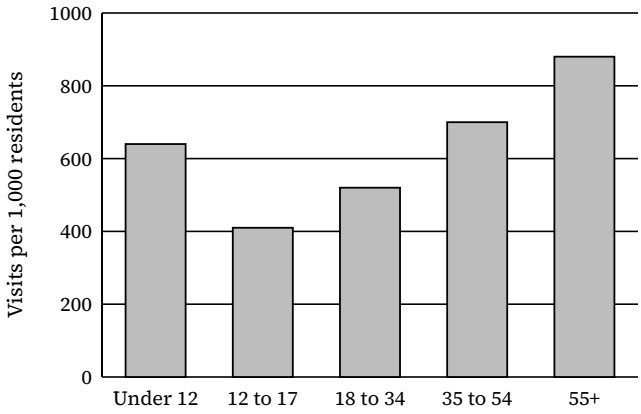
B) Restaurant openings doubled by a similar amount in neighbourhoods across the city that are far from the subway line.

C) The subway line opened later than originally planned.

D) Some riders say they use the subway to reach restaurants.

17 COMMAND OF EVIDENCE: QUANTITATIVE

Library Visits per 1,000 Residents, by Age Group, 2023



The Millbrook Public Library recently expanded its weekend programs. Staff reviewed visit data for 2023 and noted that although residents aged 55 and over made the most visits per 1,000 residents, with about 880, the group making the fewest visits was _____.

Which choice most effectively uses data from the graph to complete the text?

(A) residents aged 18 to 34, with about 520.

(B) residents aged 12 to 17, with about 410.

(C) residents under 12, with about 640.

(D) residents aged 35 to 54, with about 700.

18 COMMAND OF EVIDENCE: QUANTITATIVE

Average Ratings of Three Apple Varieties (scale of 1 to 10)

Variety	Sweetness	Crispness
Fuji	8.1	7.4
Gala	7.2	6.8
Granny Smith	4.6	8.3

A food science class asked 60 students to rate three apple varieties for sweetness and crispness on a scale from 1 (lowest) to 10 (highest). The teacher claims that sweetness and crispness do not always rise together, noting that _____.

Which choice most effectively uses data from the table to complete the statement?

(A) Granny Smith apples received the lowest average sweetness rating but the highest average crispness rating.

(B) Gala apples received an average sweetness rating of 7.2.

(C) Fuji apples received higher ratings than Gala apples on both measures.

(D) all three varieties received an average crispness rating above 6.

19 BOUNDARIES

The 1969 lunar landing relied on computers far less powerful than a _____ made up for the gap with careful, repeated testing of every procedure.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) modern phone, engineers
B) modern phone engineers
C) modern phone; engineers
D) modern phone; and engineers

20 FORM, STRUCTURE, AND SENSE

The collection of letters written by soldiers during the war _____ now held at the university library.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) are
B) is
C) were
D) have been

21 FORM, STRUCTURE, AND SENSE

By the time the festival ended on Sunday evening, volunteers _____ more than two tonnes of litter from the park.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) collected
B) have collected
C) collect
D) had collected

22 FORM, STRUCTURE, AND SENSE

The review boards of both towns submitted plans, but the two _____ proposals were rejected for exceeding the budget.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) boards
B) boards'
C) board's
D) boards's

23 FORM, STRUCTURE, AND SENSE

Walking along the shoreline at dawn, _____

Which choice completes the text so that it conforms to the conventions of Standard English?

A) we startled a flock of sandpipers.

B) a flock of sandpipers rose as we approached.

C) the shoreline was crowded with sandpipers.

D) sandpipers were startled by our arrival.

24 BOUNDARIES

Botanist Henrik Lund's study of the edelweiss _____ found that it flowers earlier each decade.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) edelweiss, a plant native to alpine meadows,

B) edelweiss a plant native to alpine meadows,

C) edelweiss, a plant native to alpine meadows

D) edelweiss; a plant native to alpine meadows,

25 BOUNDARIES

The expedition's equipment included three _____ a satellite phone, a water purifier, and a first-aid kit.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) essentials,

B) essentials;

C) essentials:

D) essentials

26 FORM, STRUCTURE, AND SENSE

The orchestra celebrated _____ hundredth anniversary with a free concert in the park.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) its

B) it's

C) its'

D) their's

Questions 27 to 29 of 33

27 TRANSITIONS

Elena's first attempt at a loaf of sourdough collapsed in the oven. _____ she kept a careful log of every change to the recipe, and within a month her loaves were rising evenly.

Which choice completes the text with the most logical transition?

A) For example,

B) As a result,

C) Similarly,

D) Nevertheless,

28 TRANSITIONS

Many cities have added bike lanes to reduce traffic. _____ the lanes improve air quality, since fewer cars on the road means fewer exhaust emissions.

Which choice completes the text with the most logical transition?

A) However,

B) Moreover,

C) Instead,

D) Previously,

29 TRANSITIONS

The first telescopes magnified distant objects, but their lenses bent different colours of light by different amounts, blurring the images. _____ later designs used mirrors, which reflect all colours equally.

Which choice completes the text with the most logical transition?

A) To address this problem,

B) Likewise,

C) For instance,

D) In summary,

30 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- The Great Barrier Reef lies off the coast of Queensland, Australia.
- It stretches for about 2,300 kilometres.
- It is made up of nearly 3,000 individual reefs.
- It was named a World Heritage Site in 1981.

The student wants to emphasize the reef's size. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The Great Barrier Reef was named a World Heritage Site in 1981.

(B) The Great Barrier Reef lies off the coast of Queensland, Australia.

(C) Stretching about 2,300 kilometres off Australia's coast, the Great Barrier Reef is made up of nearly 3,000 individual reefs.

(D) Australia's Great Barrier Reef is one of the best-known natural sites in the world.

31 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- Marisol Quiñones is an urban planner.
- Her 2021 book is titled *Green Streets*.
- The book examines how trees affect city neighbourhoods.
- It reports that tree-lined streets can lower summer temperatures by several degrees.

The student wants to introduce the book to an audience already familiar with Quiñones. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) *Green Streets*, Quiñones's 2021 book, examines how tree-lined streets can lower summer temperatures in city neighbourhoods.

(B) Marisol Quiñones, an urban planner, has written books about cities.

(C) Tree-lined streets can lower summer temperatures by several degrees.

(D) A book published in 2021 argues that cities should plant more trees.

32 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- Ada Lovelace lived from 1815 to 1852.
- In 1843 she published notes on Charles Babbage's Analytical Engine, a proposed mechanical computer.
- The notes included a step-by-step method for the engine to calculate a sequence of numbers.
- Many historians consider this method the first published computer program.

The student wants to explain the significance of Lovelace's notes to an audience unfamiliar with her.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Ada Lovelace lived from 1815 to 1852.

(B) Lovelace published notes in 1843 about a proposed mechanical computer.

(C) Lovelace's 1843 notes on Babbage's proposed Analytical Engine included a step-by-step method that many historians consider the first published computer program.

(D) The Analytical Engine was designed to calculate sequences of numbers.

33 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- The Brooklyn Bridge opened in 1883.
- Its main span is about 1,595 feet long.
- The Golden Gate Bridge opened in 1937.
- Its main span is 4,200 feet long.
- Both are suspension bridges.

The student wants to emphasize a difference in the size of the two bridges. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Both the Brooklyn Bridge and the Golden Gate Bridge are suspension bridges.

(B) The Brooklyn Bridge opened in 1883, more than fifty years before the Golden Gate Bridge.

(C) The Brooklyn Bridge and the Golden Gate Bridge are both famous landmarks.

(D) The Golden Gate Bridge's 4,200-foot main span is more than two and a half times as long as the Brooklyn Bridge's span of about 1,595 feet.

STOP If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

1 Reading and Writing Module 2

33 QUESTIONS 39 MINUTES

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer. Circle your answer in the booklet.

PACE YOURSELF

You have **39 minutes** for **33 questions**, which is about **71 seconds** a question. Answer what you can quickly, circle the ones that need more time, and come back to them before time is called. There is no penalty for a wrong answer, so never leave a question blank. This module is harder than Module 1.

WHAT YOU WILL SEE IN THIS MODULE

- Words in context
- Text structure and purpose
- Cross-text connections
- Central ideas and details
- Command of evidence: quantitative
- Command of evidence: textual
- Inferences
- Form, structure, and sense
- Boundaries
- Transitions
- Rhetorical synthesis

Questions 1 to 4 of 33

1 WORDS IN CONTEXT

Despite the committee's _____ praise for the architect's early sketches, its final report was far more guarded, noting that several of the proposed materials had never been tested in so exposed a coastal setting.

Which choice completes the text with the most logical and precise word or phrase?

(A) tepid
(B) effusive
(C) grudging
(D) perfunctory

2 WORDS IN CONTEXT

Historians have long treated the 1888 diary as a _____ source, quoting it freely; only recently did a paper-conservation team show that several pages had been rewritten decades later, and the diary's authority has since been questioned.

Which choice completes the text with the most logical and precise word or phrase?

(A) controversial
(B) obsolete
(C) negligible
(D) reliable

3 WORDS IN CONTEXT

Botanist Imani Okafor's field notes are notable for their _____ : she records not only the height and leaf shape of each specimen but also the soil temperature, the slope of the ground and the hour at which the sample was collected.

Which choice completes the text with the most logical and precise word or phrase?

(A) thoroughness
(B) brevity
(C) ambiguity
(D) informality

4 WORDS IN CONTEXT

Because the lake's chemistry shifts so little from one season to the next, researchers consider it unusually _____ ; even a severe drought left its mineral balance almost untouched.

Which choice completes the text with the most logical and precise word or phrase?

(A) volatile
(B) porous
(C) stable
(D) scarce

5 WORDS IN CONTEXT

The critic's review was not hostile, exactly, but it was _____ : every compliment was followed by a qualification so lengthy that the reader could scarcely remember what had been praised.

Which choice completes the text with the most logical and precise word or phrase?

(A) rapturous

(B) equivocal

(C) unflinching

(D) incidental

6 TEXT STRUCTURE AND PURPOSE

The following text is adapted from an original essay about the history of cartography.

> Early mapmakers rarely admitted what they did not know. Where a coastline had not been surveyed, they did not leave the page blank; they drew sea monsters, or a confident but invented shore. Such flourishes should not be dismissed as mere decoration. They show how strongly a map was expected to look complete, and how reluctant its makers were to let the viewer see the edge of their knowledge.

Which choice best describes the function of the underlined sentence in the text as a whole?

(A) It concedes that a feature just described had no purpose beyond ornament.

(B) It introduces a counterexample to the claim made in the first sentence.

(C) It summarises the methods that mapmakers used to survey coastlines.

(D) It argues that a feature just described reveals something about the attitudes of the people who made it.

7 TEXT STRUCTURE AND PURPOSE

The following text is from an original short story. Ansel is preparing to leave the village where he has lived all his life.

> Ansel had expected the packing to take all night. Instead it took an hour: a coat, two books, his grandfather's razor wrapped in a shirt. He sat on the bare cot and looked at the room, which now looked no different from the day he had been born into it, and wondered why it should feel like an accusation that so little of his life could be carried. In the morning he left the key on the table, as though someone might come back for it.

Which choice best states the main purpose of the text?

(A) To explain why Ansel has chosen to leave the village

(B) To contrast Ansel's careful planning with his sudden change of mind

(C) To convey Ansel's uneasy sense that his departure has exposed how little he can claim as his own

(D) To show that Ansel's belongings are valuable to other people

8 TEXT STRUCTURE AND PURPOSE

The following text is from an original poem.

> I never learned the names of stars.
> My father knew them, pointing out
> a ladle, hunter, swan, the scars
> of light that left no room for doubt.
> Now when I tilt my head, I find
> no pattern, only scattered grain;
> yet something in me, left behind,
> still turns toward them like a vane.

Which choice best describes the overall structure of the text?

(A) The speaker describes a former source of certainty, then admits to a loss before noting a lingering instinct.

(B) The speaker argues with a claim, then concedes that it was correct.

(C) The speaker tells a story in order, then draws a general moral from it.

(D) The speaker describes a place, then compares it with an unfamiliar one.

9 CROSS-TEXT CONNECTIONS

TEXT 1

Some linguists maintain that a language with few speakers is endangered chiefly because its young people see little economic advantage in learning it. On this view, the most effective protection is to make the language useful: to teach it in schools and to give it a role in commerce and government.

TEXT 2

A sociolinguist who has studied communities in which a small language survived notes that economic incentives were rarely the deciding factor. In those communities, children kept the language because it was spoken at home, and the strongest predictor of survival was whether grandparents and grandchildren talked to one another.

Based on the texts, how would the sociolinguist in Text 2 most likely respond to the view described in Text 1?

(A) By agreeing that a language survives only if it has commercial value

(B) By arguing that schools should never teach small languages

(C) By claiming that communities with few speakers cannot protect their language

(D) By suggesting that the view gives too little weight to the part played by family life in keeping a language alive

10 CENTRAL IDEAS AND DETAILS

The following text is adapted from an original science article.

> For decades, biologists assumed that the pale colour of cave-dwelling fish was simply the absence of something: without light, the pigment-producing machinery had no use and wore away. Recent genetic work suggests otherwise. In several species the genes for pigment remain intact but are switched off, apparently because making pigment is metabolically costly, and a fish that saves energy in a food-poor cave gains a small but real advantage over one that does not.

Which choice best states the main idea of the text?

(A) Cave fish have lost the genes needed to produce pigment because they live in darkness.

(B) A long-held explanation for the colour of cave fish is being replaced by one in which the loss of pigment is itself an adaptation.

(C) Cave fish are easier to study than fish that live in open water.

(D) Pigment production is the most expensive process in the bodies of fish.

11 CENTRAL IDEAS AND DETAILS

The following text is from an original essay about public libraries.

> A library is often judged by its circulation figures, but these measure only one thing: books leaving the building. They say nothing of the retiree who comes to read the newspaper, the teenager who finishes homework at a table with a working outlet, or the new arrival who learns English from a volunteer in the corner. A library may lend fewer books than it did a generation ago and yet be busier, and more needed, than ever.

Which choice best states the main idea of the text?

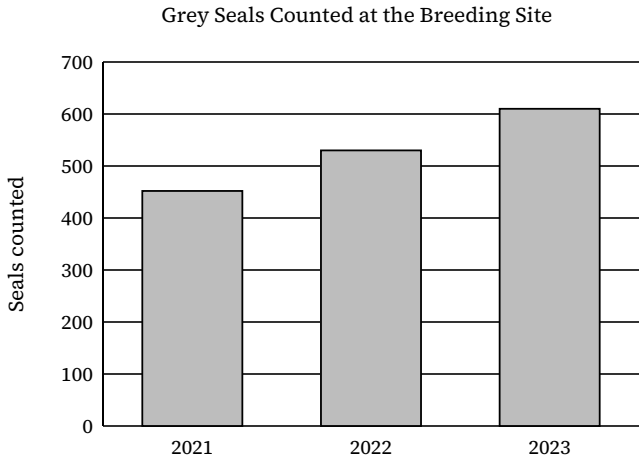
(A) Circulation figures capture too narrow a part of what libraries do for the people who use them.

(B) Libraries should stop lending books and offer other services instead.

(C) Most library visitors come to read newspapers and magazines.

(D) Libraries are less busy than they were a generation ago.

12 COMMAND OF EVIDENCE: QUANTITATIVE



A marine biologist recorded the number of grey seals counted at a breeding site in five consecutive years. In 2019 the site held 410 seals and in 2020 it held 438; the counts for the later years are shown in the graph.

Which choice most effectively uses data from the graph to complete the text?

(A) Between 2021 and 2023 the count rose slightly, from 452 to 610, an increase of about 15 percent.

(B) Between 2021 and 2023 the count fell, from 610 to 452, a decrease of about 35 percent.

(C) Between 2021 and 2023 the count rose sharply, from 452 to 610, an increase of about 35 percent.

(D) Between 2020 and 2022 the count rose sharply, from 438 to 570, an increase of about 35 percent.

13 COMMAND OF EVIDENCE: QUANTITATIVE

Percent of Surveyed Students Who Cycle to Campus

Distance from campus	2018	2023
Within 5 kilometres	24%	31%
More than 5 kilometres	6%	6%

A university surveyed 600 students on how they usually travel to campus. A researcher wants to claim that the share of students who cycle is rising only among those who live within five kilometres.

Which choice most effectively uses data from the table to support the researcher's claim?

A. Among students living within five kilometres, the share who cycle rose from 24 percent to 31 percent, and among those living farther away it rose from 6 percent to 9 percent.

B. Among students living farther away, the share who cycle stayed at 6 percent, while among those living within five kilometres it fell from 31 percent to 24 percent.

C. Overall, 31 percent of students living within five kilometres cycle, compared with 6 percent of those living farther away.

D. Among students living within five kilometres, the share who cycle rose from 24 percent to 31 percent, while among those living farther away it stayed at 6 percent.

14 COMMAND OF EVIDENCE: TEXTUAL

The following text is adapted from an original essay about a playwright.

> Critics have often described the playwright Ilse Brandt as a miniaturist, a writer of small rooms and quiet quarrels. Yet her late plays, set in occupied cities, abandon the domestic scale altogether. In one, a single scene brings together forty characters, none of whom speaks to another for more than a line.

A student claims that Brandt's late plays differ from the work for which she was first known. Which quotation from the text best supports the student's claim?

A. "her late plays, set in occupied cities, abandon the domestic scale altogether"

B. "Critics have often described the playwright Ilse Brandt as a miniaturist"

C. "a writer of small rooms and quiet quarrels"

D. "none of whom speaks to another for more than a line"

15 COMMAND OF EVIDENCE: TEXTUAL

A researcher claims that the migratory birds known as bar-tailed godwits can fly without rest for more than a week.

> In 2022, researchers fitted a godwit with a tracker and followed it from Alaska to Tasmania. The bird reached its destination in eleven days, and the tracker recorded no stops: no landing on any island or coast along the way.

Which finding, if true, would most directly support the researcher's claim?

(A) The bird travelled farther than any other bird the researchers had tracked.

(B) The bird was heavier when it departed than when it reached Tasmania.

(C) The tracker recorded the bird flying for eleven days without landing.

(D) Other birds fitted with trackers followed the same route.

16 COMMAND OF EVIDENCE: TEXTUAL

The following text is adapted from an original essay on translation.

> A translator, the novelist argued, is less a window than a lens: the reader never sees the original text unaltered, but sees it bent in a particular direction by the translator's choices. This is not a flaw to be regretted. A window pretends the glass is not there; a lens admits it and, in doing so, lets the reader judge what has been bent.

Which quotation from the text best illustrates the novelist's view that translation should be open about the translator's role?

(A) "A translator, the novelist argued, is less a window than a lens"

(B) "A window pretends the glass is not there; a lens admits it and, in doing so, lets the reader judge what has been bent."

(C) "the reader never sees the original text unaltered"

(D) "This is not a flaw to be regretted."

17 INFERENCES

The following text is adapted from an original science article.

> When the first reliable thermometers came into use, observers in northern towns began to keep daily temperature records. These early records are valuable, but a historian of science cautions that most were kept by the owners of instruments placed against sunlit house walls, and that a thermometer kept in such a spot reads warmer than the air around it. It follows that

Which choice most logically completes the text?

(A) modern thermometers are no more accurate than the earliest ones.

(B) the owners of the instruments deliberately falsified their readings.

(C) temperature records from the earliest instruments may overstate how warm the period was.

(D) daily records were rarely kept in northern towns.

18 INFERENCES

The following text is from an original essay about city planning.

> Planners once believed that wider roads would relieve congestion. Yet studies of cities that widened their main roads found that traffic soon grew to fill the new space, because the easier journey encouraged more people to drive. If this pattern is typical, then a city that hopes to reduce congestion would probably

Which choice most logically completes the text?

(A) gain little by studying the traffic of other cities.

(B) achieve more by offering alternatives to driving than by adding road capacity.

(C) reduce congestion by removing all of its main roads.

(D) find that wider roads gradually lower the number of drivers.

19 INFERENCES

The following text is adapted from an original essay about a violinist.

> The violinist Perrine Aubert is known for playing the same pieces in very different ways from one concert to the next. Reviewers who heard her two performances of a sonata a week apart reported almost opposite impressions, one calling it austere and the other lavish. Aubert herself says she does not decide on an interpretation beforehand. Her approach suggests that for her, a performance is

Which choice most logically completes the text?

(A) something that takes shape in the moment rather than a reproduction of a prepared plan.

(B) a test of how closely she can imitate other violinists.

(C) a sequence of steps she rehearses until they are fixed.

(D) less important to her than the reviews that follow it.

20 FORM, STRUCTURE, AND SENSE

The ceramics of the Jōmon period, some of the oldest known, _____ by cord-like patterns pressed into the wet clay.

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) is distinguished

(B) distinguishes

(C) having been distinguished

(D) are distinguished

21 BOUNDARIES

Neither the architect nor the engineers _____ convinced that the original design could bear the additional load.

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) was

(B) is

(C) has been

(D) were

22 FORM, STRUCTURE, AND SENSE

The glaciologist spent the winter measuring the ice sheet, _____ that the thinning she had suspected was far more rapid than any model had predicted.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) discovering
B) discovered
C) she discovered
D) and discovering

23 FORM, STRUCTURE, AND SENSE

Of the two proposed routes for the new railway, the coastal line, which would cost _____ the inland one, was ultimately chosen.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) least than
B) less than
C) the less of
D) fewer as

24 BOUNDARIES

The scientist who led the expedition, along with her colleagues from three universities, _____ the findings at the conference in March.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) were presenting
B) have presented
C) presented
D) present

25 BOUNDARIES

Marguerite, _____ grandmother had run the bakery for forty years, decided to keep the recipes exactly as they were.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) who's
B) whom
C) which
D) whose

26 FORM, STRUCTURE, AND SENSE

When the committee reviewed the budget, it found that spending on maintenance had _____ by nearly a third over five years, while spending on new equipment had barely changed.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) risen
B) rose
C) raised
D) rised

27 TRANSITIONS

Sea-level rise is often described as a slow, uniform process. In practice, however, it is uneven; some coasts, where the land itself is sinking, face a rise far steeper than the global average. _____, the same ocean can pose very different risks to neighbouring cities.

Which choice completes the text with the most logical transition?

A) Instead
B) Likewise
C) Consequently
D) Previously

28 TRANSITIONS

Early reviewers of the novel dismissed its fragmented structure as a failure of craft. Later critics, _____, came to see the fragmentation as the book's central achievement, a form that mirrors the disordered memory of its narrator.

Which choice completes the text with the most logical transition?

A) for example
B) by contrast
C) in other words
D) as a result

29 TRANSITIONS

Few musical instruments are as widely studied as the violin. _____ the lute, which was once equally popular, has received comparatively little attention from scientists and historians alike.

Which choice completes the text with the most logical transition?

A) By contrast,
B) For this reason,
C) Similarly,
D) In addition,

30 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- The Hallam Mill is a nineteenth-century cotton mill in northern England.
- It closed in 1974 and stood empty for decades.
- A restoration project began in 2016.
- The project involved replacing more than 4,000 square metres of roof.
- It was completed in 2021 at a cost of about 18 million pounds.

The student wants to emphasise the scale of the restoration project.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The Hallam Mill, a cotton mill in northern England, closed in 1974 and stood empty for decades.

(B) The restoration of the Hallam Mill, which began in 2016 and cost about 18 million pounds, involved replacing more than 4,000 square metres of roof.

(C) Restoration of the Hallam Mill, a nineteenth-century cotton mill, began in 2016 and was completed in 2021.

(D) The Hallam Mill, which stood empty after 1974, was restored over five years.

31 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- The book is titled The Salt Road.
- It was published in 2020.
- Its author, Wen Zhao, is a historian of trade.
- Zhao spent six years tracing ancient salt routes across Central Asia.
- The book was shortlisted for a national prize for non-fiction.

The student wants to introduce a book to readers who do not know its author.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The Salt Road, published in 2020, was shortlisted for a national prize for non-fiction.

(B) The Salt Road is a book about salt routes that took six years to research.

(C) Published in 2020, The Salt Road is a work of non-fiction about trade and salt.

(D) The Salt Road (2020) is by Wen Zhao, a historian of trade who spent six years tracing ancient salt routes across Central Asia.

32 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- Crop rotation means growing different crops in the same field in successive seasons.
- Planting beans one year and grain the next is a common pattern.
- Beans add nitrogen to the soil.
- Grain crops use up nitrogen.
- Fields rotated this way often need less fertiliser.

The student wants to explain the benefit of an old farming practice.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Crop rotation, in which different crops are grown in a field in successive seasons, is a common farming practice.

(B) Planting beans one year and grain the next is a common pattern of crop rotation.

(C) Because beans add nitrogen to the soil that grain crops use up, rotating the two often reduces the fertiliser a field needs.

(D) Grain crops use up nitrogen, while beans add it to the soil.

33 TRANSITIONS

The settlers had expected to find the valley fertile and well watered. _____ , the first summer brought a drought that killed most of their seedlings.

Which choice completes the text with the most logical transition?

(A) Instead

(B) Moreover

(C) Therefore

(D) In particular

STOP If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

2 Math

27 QUESTIONS 43 MINUTES

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions.

NOTES

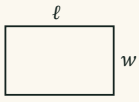
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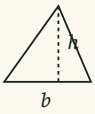
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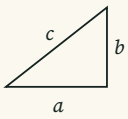
$A = \pi r^2$
$C = 2\pi r$



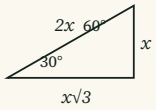
$A = \ell w$

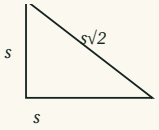


$A = \frac{1}{2}bh$

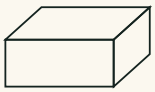


$c^2 = a^2 + b^2$

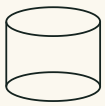




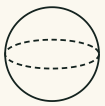
Special right triangles



$V = \ell wh$



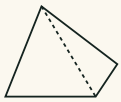
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360. The number of radians of arc in a circle is 2π. The sum of the measures in degrees of the angles of a triangle is 180.

ANSWERING THE QUESTIONS

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in the booklet. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer in the box under the question.

- If you find **more than one correct answer,** write only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long, write the decimal equivalent.
- If your answer is a **decimal** that is too long, truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as 3½), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Do not include **symbols** such as a percent sign, comma, or dollar sign in your answer.

PACE YOURSELF

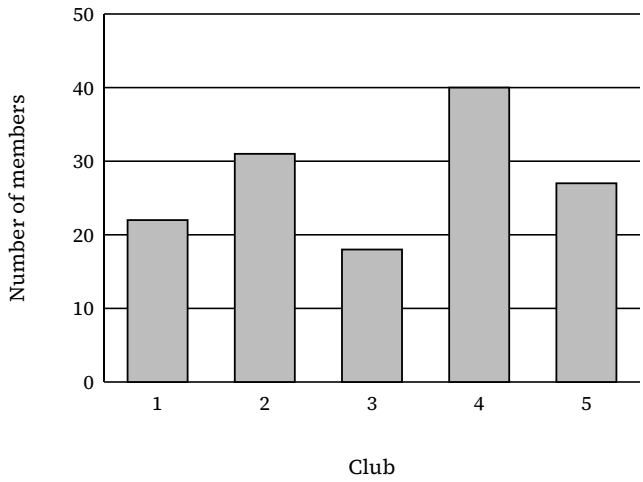
You have **43 minutes** for **27 questions**, which is about **96 seconds** a question. Answer what you can quickly, circle the ones that need more time, and come back to them before time is called. There is no penalty for a wrong answer, so never leave a question blank.

WHAT YOU WILL SEE IN THIS MODULE

Data analysis | Percentages | Linear equations | Linear equations in context | Systems of equations | Functions | Nonlinear expressions | Ratios and rates | Triangles | Circles | Quadratics | Probability | Statistics | Scatterplots | Inequalities | Exponential functions | Systems of equations in context | Radicals | Linear functions | Area | Percent change | Volume | Right triangles

Questions 1 to 5 of 27

1 DATA ANALYSIS



The bar graph shows the number of members in each of five clubs. How many more members does Club 4 have than Club 3?

(A) 12

(B) 22

(C) 24

(D) 58

2 PERCENTAGES

What is 35% of 240?

(A) 70

(B) 84

(C) 105

(D) 205

3 LINEAR EQUATIONS

If $5x - 12 = 3x + 10$, what is the value of x?

(A) 5

(B) 9

(C) 11

(D) 22

4 LINEAR EQUATIONS IN CONTEXT

A gym charges a one-time sign-up fee of $30 plus $18 per month. Which equation gives the total cost y, in dollars, of a membership for m months?

(A) $y = 18m + 30$

(B) $y = 30m + 18$

(C) $y = 48m$

(D) $y = 18m - 30$

5 SYSTEMS OF EQUATIONS

$x + y = 14$
$x - y = 6$

What is the value of x?

(A) 4

(B) 8

(C) 10

(D) 20

6 FUNCTIONS

x	1	2	3	4
f(x)	7	10	13	16

The table shows four values of the linear function f. Which equation defines f?

A $f(x) = 4x + 3$

B $f(x) = 3x + 7$

C $f(x) = 7x$

D $f(x) = 3x + 4$

7 NONLINEAR EXPRESSIONS

Which expression is equivalent to $(2x^3)(5x^4)$?

A $7x^7$

B $10x^7$

C $10x^{12}$

D $7x^{12}$

8 RATIOS AND RATES

A recipe uses 3 cups of flour for every 2 cups of sugar. How many cups of flour are needed for 10 cups of sugar?

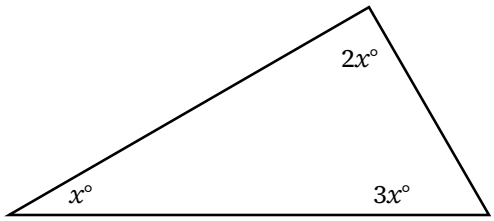
A 15

B 12

C 18

D 20

9 TRIANGLES



In the triangle shown, the three angles measure $x°$, $2x°$, and $3x°$. What is the value of x?

A 20

B 45

C 30

D 60

10 CIRCLES

A circle has a circumference of 12π centimetres. What is the area of the circle, in square centimetres?

A 12π

B 36π

C 6π

D 144π

11 QUADRATICS

What are the solutions to $x^2 - 5x - 14 = 0$?

A) −2 and 7

B) 2 and 7

C) −7 and 2

D) −2 and −7

12 PROBABILITY

How Students Travel to School

	Walk	Bike	Bus
Grade 9	18	12	20
Grade 10	14	16	20

The table shows how 100 students travel to school. If one student who takes the bus is chosen at random, what is the probability that the student is in Grade 10?

A) $\frac{1}{5}$

B) $\frac{2}{5}$

C) $\frac{3}{5}$

D) $\frac{1}{2}$

13 STATISTICS

The mean of five test scores is 80. A sixth score of 92 is added. What is the mean of all six scores?

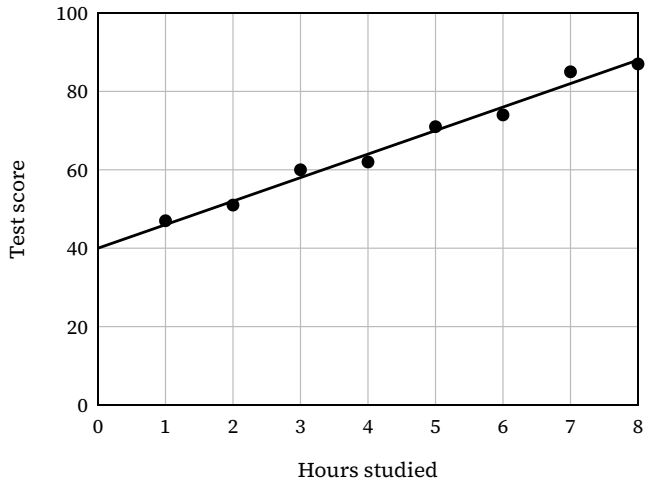
A) 81

B) 82

C) 83

D) 86

14 SCATTERPLOTS



The scatterplot shows the hours students studied and their test scores. The line of best fit is $y = 6x + 40$. According to the line, what is the predicted score for a student who studies 7 hours?

A) 52

B) 76

C) 82

D) 88

15 INEQUALITIES

A school needs at least 120 volunteers, and 45 have already signed up. If 15 new volunteers sign up each week, which inequality represents the number of weeks, w, needed to reach the goal?

A) $45 + 15w \geq 120$

B) $45 + 15w \leq 120$

C) $15 + 45w \geq 120$

D) $120 + 15w \geq 45$

16 FUNCTIONS

The function f is defined by $f(x) = 2x^2 - 3x + 1$. What is the value of $f(3)$?

A) 4

B) 16

C) 22

D) 10

17 EXPONENTIAL FUNCTIONS

A population of bacteria doubles every 3 hours. There are 500 bacteria at the start. Which function gives the number of bacteria after t hours?

A) $P(t) = 500(2)^{3t}$

B) $P(t) = 500(2)^{t/3}$

C) $P(t) = 500(3)^{t/2}$

D) $P(t) = 1{,}000(t/3)$

18 SYSTEMS OF EQUATIONS IN CONTEXT

A theatre sold 120 tickets. Adult tickets cost \$12 and child tickets cost \$7, and total ticket sales were \$1,090. How many adult tickets were sold?

A) 40

B) 50

C) 60

D) 70

19 RADICALS

If $\sqrt{(x + 7)} = 5$, what is the value of x?

A) 12

B) 25

C) 18

D) 32

20 LINEAR FUNCTIONS

A line in the xy-plane passes through the points (1, 3) and (4, 9). Which equation represents the line?

A) $y = 2x + 1$

B) $y = 3x$

C) $y = 2x + 3$

D) $y = x + 2$

Questions 21 to 27 of 27

21 LINEAR EQUATIONS

If $\frac{3}{4}$ of a number is 21, what is the number?

Write your answer

22 AREA

What is the area, in square units, of a triangle with a base of 14 and a height of 9?

Write your answer

23 LINEAR EQUATIONS

If $2(x - 3) = 3x - 11$, what is the value of x?

Write your answer

24 PERCENT CHANGE

The price of a concert ticket rose from \$40 to \$52. By what percent did the price increase?

Write your answer

25 QUADRATICS

What is the positive solution to the equation $x^2 - 6x - 16 = 0$?

Write your answer

26 VOLUME

A cylinder has a radius of 3 and a height of 10. Its volume is $k\pi$. What is the value of k?

Write your answer

27 RIGHT TRIANGLES

In a right triangle, one leg has length 5 and the hypotenuse has length 13. What is the length of the other leg?

Write your answer

STOP If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

2 Math Module 2

27 QUESTIONS 43 MINUTES

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions.

NOTES

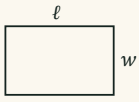
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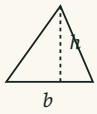
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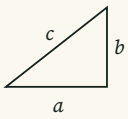
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$C = 2\pi r$



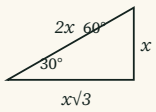
$A = \ell w$

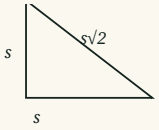


$A = \frac{1}{2}bh$

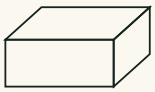


$c^2 = a^2 + b^2$

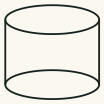




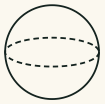
Special right triangles



$V = \ell wh$



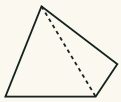
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360. The number of radians of arc in a circle is 2π. The sum of the measures in degrees of the angles of a triangle is 180.

Questions 1 to 6 of 27

1 FUNCTIONS

The function f is defined by $f(x) = 3x^2 - 8x + 4$. What is the value of $f(4)$?

A) 84

B) −16

C) 24

D) 20

2 SYSTEMS OF EQUATIONS

$x + y = 12$
$x - y = 8$

What is the value of x?

A) 10

B) 2

C) 12

D) 8

3 QUADRATICS

What are the solutions to $x^2 + 2x - 48 = 0$?

A) 6 and 8

B) −6 and −8

C) −8 and 6

D) −6 and 8

4 PERCENTAGES

What is 20% of 160?

A) 128

B) 32

C) 64

D) 180

5 EXPONENTIAL FUNCTIONS

A population of bacteria doubles every 6 hours. There are 300 bacteria at the start. Which function gives the number of bacteria after t hours?

A) $P(t) = 300(2)^{6t}$

B) $P(t) = 300(6)^{t/2}$

C) $P(t) = 600(t/6)$

D) $P(t) = 300(2)^{t/6}$

6 LINEAR FUNCTIONS

A line in the xy-plane passes through the points (1, 7) and (4, 16). Which equation represents the line?

A) $y = 4x + 4$

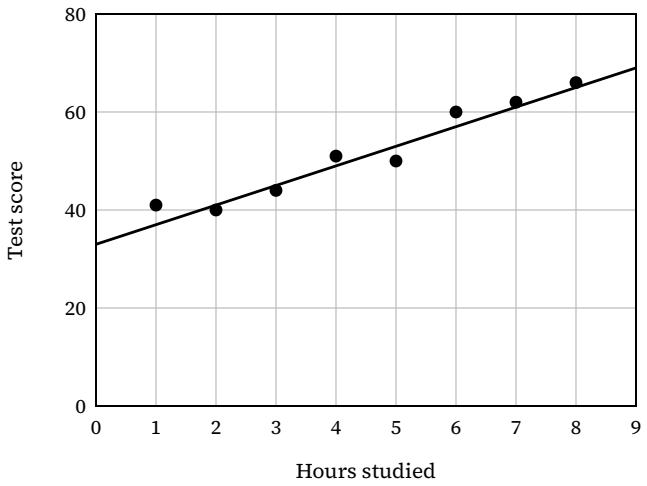
B) $y = 3x + 6$

C) $y = 3x + 4$

D) $y = 2x - 4$

7

SCATTERPLOTS



The scatterplot shows the hours students studied and their test scores. The line of best fit is $y = 4x + 33$. According to the line, what is the predicted score for a student who studies 8 hours?

A) 65

B) 32

C) 61

D) 45

8

CIRCLES IN THE PLANE

The equation $(x - 4)^2 + (y + 1)^2 = 25$ represents circle A in the xy-plane. Circle B is obtained by shifting circle A 4 units left and 3 units down. Which of the following equations represents circle B?

A) $(x - 8)^2 + (y + 4)^2 = 25$

B) $x^2 + (y + 4)^2 = 25$

C) $x^2 + (y - 2)^2 = 25$

D) $(x - 8)^2 + (y - 2)^2 = 25$

9

QUADRATICS

The function f is defined by $f(x) = x^2 - 10x + 22$. What is the minimum value of f?

A) -3

B) 3

C) 22

D) 5

10

FUNCTIONS

The functions f and g are defined by $f(x) = 4x + 1$ and $g(x) = x^2 + 4$. What is the value of $f(g(3)) - g(f(3))$?

A) 226

B) 169

C) -120

D) 120

11

SYSTEMS OF EQUATIONS

$ax + 3y = 13$
$16x + 12y = 49$

In the given system of equations, a is a constant. If the system has no solution, what is the value of a?

A) 16

B) 8

C) 12

D) 4

12 SURFACE AREA

Two identical cubes are glued together along one full face to form a rectangular solid. The surface area of the solid is 640 square centimetres. What is the edge length, in centimetres, of each cube?

A) 10

B) 8

C) 7

D) 11

13 CIRCLES

A sector of a circle has a radius of 12 and a central angle of $\frac{\pi}{4}$ radians. What is the length of the arc of the sector?

A) 3π

B) 12π

C) 36π

D) 24π

14 TRIGONOMETRY

In a right triangle, angle A is acute and $\sin A = \frac{5}{13}$. What is the value of $\tan A$?

A) $\frac{12}{13}$

B) $\frac{12}{5}$

C) $\frac{5}{12}$

D) $\frac{13}{5}$

15 PERCENTAGES

The price of an item is raised by 20%. Later, the new price is reduced by 30%. Compared with the original price, the final price is

A) 10% lower

B) 50% lower

C) the same as the original price

D) 16% lower

16 ABSOLUTE VALUE

How many integers x satisfy the inequality $|x - 8| \leq 5$?

A) 10

B) 11

C) 6

D) 5

17 EXPONENTIAL EQUATIONS

If $2^{x+1} = 4^{x-3}$, what is the value of x?

A) 4

B) 5

C) 9

D) 7

18 PERCENTAGES

70 is p% greater than 20. What is the value of p?

(A) 350

(B) 250

(C) 71

(D) 50

19 EXPONENTIAL FUNCTIONS

$V(x) = 4{,}000(0.7)^{x/6}$

The function V gives the value, in dollars, of a certain machine x months after it is purchased. By what percent does the value of the machine decrease each year?

(A) 70%

(B) 30%

(C) 51%

(D) 49%

20 STATISTICS

Data set A

Value	1	2	3	4	5
	2	6	7	9	10

Data set A has the five values shown. Data set B is formed by multiplying each value in A by 2 and then adding 8. Which of the following correctly compares the means and the standard deviations of the two data sets?

(A) The mean of data set B is 14.8 greater than the mean of data set A, and the standard deviation of data set B is 2 times the standard deviation of data set A.

(B) The mean of data set B is 14.8 greater than the mean of data set A, and the standard deviations are equal.

(C) The mean of data set B is 8 greater than the mean of data set A, and the standard deviation of data set B is 2 times the standard deviation of data set A.

(D) The mean of data set B is 2 times the mean of data set A, and the standard deviation of data set B is 8 greater than the standard deviation of data set A.

21 SYSTEMS OF EQUATIONS

The sum of two numbers is 29, and their difference is 15. What is the larger of the two numbers?

Write your answer

Questions 22 to 27 of 27

22 QUADRATICS

$3x^2 - 12x + 0 = 0$

What is the sum of the solutions to the given equation?

Write your answer

23 SIMILAR TRIANGLES

Triangle ABC is similar to triangle DEF, where A corresponds to D and B corresponds to E. In triangle ABC, AB = 3, BC = 8 and AC = 9. In triangle DEF, DE = 6. What is the perimeter of triangle DEF?

Write your answer

24 CIRCLES IN THE PLANE

$x^2 + y^2 + 12x + 10y = -45$

The equation of a circle in the xy-plane is shown. What is the radius of the circle?

Write your answer

25 QUADRATICS

The height h, in metres, of a ball t seconds after it is thrown is given by $h(t) = -t^2 + 8t + 10$. What is the maximum height, in metres, that the ball reaches?

Write your answer

26 QUADRATICS

The equation $x^2 + bx + 25 = 0$, where b is a positive constant, has exactly one real solution. What is the value of b?

Write your answer

27 SYSTEMS WITH NONLINEAR EQUATIONS

$y = x^2 + 2x + 9$
$y = 4x + k$

In the given system of equations, k is a constant. If the system has exactly one solution, what is the value of k?

Write your answer

STOP If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

Answer sheet

SAT Practice Test 1, Reading and Writing

Fill in the circle for each answer, or write your student-produced answers in the boxes. Then mark yourself with the answer key.

Module 1

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D

Module 2

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D

Answer sheet

SAT Practice Test 1, Math

Fill in the circle for each answer, or write your student-produced answers in the boxes. Then mark yourself with the answer key.

Module 1

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21				
22				
23				
24				
25				
26				
27				

Module 2

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21				
22				
23				
24				
25				
26				
27				

Answer key

SAT Practice Test 1, Reading and Writing

Module 1

NO.	ANSWER	SKILL
1	**B**	Words in context
2	**A**	Words in context
3	**C**	Words in context
4	**D**	Words in context
5	**A**	Words in context
6	**C**	Text structure and purpose
7	**D**	Text structure and purpose
8	**A**	Text structure and purpose
9	**B**	Cross-text connections
10	**A**	Central ideas and details
11	**C**	Central ideas and details
12	**A**	Inferences
13	**D**	Inferences
14	**D**	Command of evidence: textual
15	**C**	Command of evidence: textual
16	**B**	Command of evidence: textual
17	**B**	Command of evidence: quantitative
18	**A**	Command of evidence: quantitative
19	**C**	Boundaries
20	**B**	Form, structure, and sense
21	**D**	Form, structure, and sense
22	**B**	Form, structure, and sense
23	**A**	Form, structure, and sense
24	**A**	Boundaries
25	**C**	Boundaries
26	**A**	Form, structure, and sense
27	**D**	Transitions
28	**B**	Transitions
29	**A**	Transitions
30	**C**	Rhetorical synthesis
31	**A**	Rhetorical synthesis
32	**C**	Rhetorical synthesis
33	**D**	Rhetorical synthesis

Module 2

NO.	ANSWER	SKILL
1	**B**	Words in context
2	**D**	Words in context
3	**A**	Words in context
4	**C**	Words in context
5	**B**	Words in context
6	**D**	Text structure and purpose
7	**C**	Text structure and purpose
8	**A**	Text structure and purpose
9	**D**	Cross-text connections
10	**B**	Central ideas and details
11	**A**	Central ideas and details
12	**C**	Command of evidence: quantitative
13	**D**	Command of evidence: quantitative
14	**A**	Command of evidence: textual
15	**C**	Command of evidence: textual
16	**B**	Command of evidence: textual
17	**C**	Inferences
18	**B**	Inferences
19	**A**	Inferences
20	**D**	Form, structure, and sense
21	**D**	Boundaries
22	**A**	Form, structure, and sense
23	**B**	Form, structure, and sense
24	**C**	Boundaries
25	**D**	Boundaries
26	**A**	Form, structure, and sense
27	**C**	Transitions
28	**B**	Transitions
29	**A**	Transitions
30	**B**	Rhetorical synthesis
31	**D**	Rhetorical synthesis
32	**C**	Rhetorical synthesis
33	**A**	Transitions

Answer key

SAT Practice Test 1, Math

Module 1

NO.	ANSWER	SKILL
1	**B**	Data analysis
2	**B**	Percentages
3	**C**	Linear equations
4	**A**	Linear equations in context
5	**C**	Systems of equations
6	**D**	Functions
7	**B**	Nonlinear expressions
8	**A**	Ratios and rates
9	**C**	Triangles
10	**B**	Circles
11	**A**	Quadratics
12	**D**	Probability
13	**B**	Statistics
14	**C**	Scatterplots
15	**A**	Inequalities
16	**D**	Functions
17	**B**	Exponential functions
18	**B**	Systems of equations in context
19	**C**	Radicals
20	**A**	Linear functions
21	**28**	Linear equations
22	**63**	Area
23	**5**	Linear equations
24	**30**	Percent change
25	**8**	Quadratics
26	**90**	Volume
27	**12**	Right triangles

Module 2

NO.	ANSWER	SKILL
1	**D**	Functions
2	**A**	Systems of equations
3	**C**	Quadratics
4	**B**	Percentages
5	**D**	Exponential functions
6	**C**	Linear functions
7	**A**	Scatterplots
8	**B**	Circles in the plane
9	**A**	Quadratics
10	**C**	Functions
11	**D**	Systems of equations
12	**B**	Surface area
13	**A**	Circles
14	**C**	Trigonometry
15	**D**	Percentages
16	**B**	Absolute value
17	**D**	Exponential equations
18	**B**	Percentages
19	**C**	Exponential functions
20	**A**	Statistics
21	**22**	Systems of equations
22	**4**	Quadratics
23	**40**	Similar triangles
24	**4**	Circles in the plane
25	**26**	Quadratics
26	**10**	Quadratics
27	**8**	Systems with nonlinear equations

Score yourself. Count the questions you got right in each module.

Reading and Writing, Module 1: ___ of 33

Module 2: ___ of 33

Math, Module 1: ___ of 27

Module 2: ___ of 27

Answer explanations

SAT Practice Test 1, Reading and Writing, Module 1

Questions 1 to 17

1 **Choice B** Words in context

The colon explains the blank: the light draws smaller fish close. To attract is to draw something near. Repel is the reverse, and the lure is not described as copying or hiding anything.

2 **Choice A** Words in context

Readers who gave up on the book found the timelines more of a burden than a pleasure. Disorienting fits; engaging and predictable would not lead readers to abandon the book.

3 **Choice C** Words in context

The sentence contrasts decorative with something that does a job, and the colon names the job: strengthening social ties. Functional is the only choice that makes that contrast.

4 **Choice D** Words in context

Even though introduces a contrast with the cheaper option of modern pigments, so the curators kept the original colours. Preserve means to keep unchanged.

5 **Choice A** Words in context

After the colon, the mayor takes no side, which is what noncommittal means. Enthusiastic and hostile both imply a clear position.

6 **Choice C** Text structure and purpose

Odile says the job is temporary, yet the final sentence reveals she would be sorry to give it up. That tension is the point of the passage. Nothing suggests resentment, and the lamps are background, not the subject.

7 **Choice D** Text structure and purpose

The first four lines catalogue sounds. The turn at "But here" reveals comfort instead of annoyance. No past event, complaint or questions appear.

8 **Choice A** Text structure and purpose

The sentence gives a specific failure, and the next sentence begins with Since then to show that wind tunnel testing followed from it. The passage treats wind as a real threat and mentions no disagreement.

9 **Choice B** Cross-text connections

The Text 2 researcher found remote work lowered output for teams that rely on unplanned conversation, so the general claim misses those cases. The researcher does not dispute that people feel less distracted.

10 **Choice A** Central ideas and details

The passage overturns the mating explanation and replaces it with a warning function. The other choices state things the text never claims, and one reverses its findings.

11 **Choice C** Central ideas and details

Marcus expects the cafeteria to be something to endure, yet the silence he shares does not feel like loneliness. He is surprised, not resentful, and no invitation or decision to leave appears.

12 **Choice A** Inferences

If the originals depend on gestures and taste, a written record without quantities can only come close. The other choices contradict the logic of the passage.

13 **Choice D** Inferences

If the necks were held low, reaching treetops cannot have been their only job, so another purpose is likely. The analysis says nothing about diet, movement or how often bones are found.

14 **Choice D** Command of evidence: textual

Removing otters and seeing urchins rise and kelp fall tests the chain Ferrante proposes. The other findings do not connect otters to kelp through urchins.

15 **Choice C** Command of evidence: textual

Dara says she is fine, but her harder and harder placing of the chairs shows what she feels. In the other choices, feelings are stated outright or not shown at all.

16 **Choice B** Command of evidence: textual

If openings doubled everywhere, the subway is unlikely to be the cause. The other findings leave the link between the line and the openings untouched or even strengthen it.

17 **Choice B** Command of evidence: quantitative

The lowest bar is the 12 to 17 group, at about 410. The other choices name groups with higher bars.

Answer explanations

SAT Practice Test 1, Reading and Writing, Module 1

Questions 18 to 33

18 Choice A Command of evidence: quantitative

Granny Smith is lowest for sweetness yet highest for crispness, so the two do not move together. The other choices either do not compare the two ratings or show them rising together.

19 Choice C Boundaries

Two complete sentences cannot be joined by a comma alone or with no punctuation. A semicolon joins them correctly. A semicolon followed by and is not standard.

20 Choice B Form, structure, and sense

The subject is the singular collection, not the plural word soldiers in the phrase after it, so the verb must be singular and present: is.

21 Choice D Form, structure, and sense

The collecting was completed before another past event (the festival ending), which calls for the past perfect: had collected.

22 Choice B Form, structure, and sense

The proposals belong to the two boards, a plural noun, so the plural possessive boards' is needed. Board's would belong to a single board.

23 Choice A Form, structure, and sense

An opening phrase must be followed by the person or thing doing the walking. Only we can walk along the shoreline; the other choices leave the phrase attached to the wrong noun.

24 Choice A Boundaries

The phrase a plant native to alpine meadows is extra information about the edelweiss, so it must be set off by a pair of commas, one before and one after.

25 Choice C Boundaries

A complete sentence followed by a list that explains it takes a colon. A comma or no mark would blur the boundary, and a semicolon is used between independent clauses.

26 Choice A Form, structure, and sense

The possessive pronoun its needs no apostrophe. It's means it is, and the other forms are not standard.

27 Choice D Transitions

The failure would normally discourage her, but she kept going, which is a contrast. Nevertheless signals that contrast.

28 Choice B Transitions

The second sentence adds a further benefit to the first, so an additive transition such as Moreover fits. However and Instead would signal a contrast.

29 Choice A Transitions

Mirrors were introduced in response to the blurring described in the first sentence. To address this problem links the two ideas.

30 Choice C Rhetorical synthesis

The goal is size, and the length and the count of individual reefs are the size facts in the notes. The other choices give a date, a location, or a claim not in the notes.

31 Choice A Rhetorical synthesis

The audience already knows the author, so the best choice names the book and says what it covers. The others either re-introduce the author or leave out the title.

32 Choice C Rhetorical synthesis

The goal is significance, and only the correct choice says why the notes matter: they contain what many consider the first published computer program. The others give dates or a bare description.

33 Choice D Rhetorical synthesis

Only the first choice compares the lengths of the main spans, which is the size difference. One choice compares opening dates, and the others state similarities.

Answer explanations

SAT Practice Test 1, Reading and Writing, Module 2

Questions 1 to 17

1 **Choice B** Words in context

The word despite signals a contrast with the guarded final report, so the earlier praise must have been warm and unrestrained. Effusive means that. The other three describe praise that is weak or reluctant, which would not contrast with a cautious report.

2 **Choice D** Words in context

The historians quoted the diary freely and its authority was questioned only after the discovery, so they had treated it as trustworthy. Reliable fits. A source thought controversial or negligible would not have been quoted freely.

3 **Choice A** Words in context

The colon introduces a long list of details recorded for each specimen, which shows thoroughness. Brevity is the opposite, and nothing in the list suggests ambiguity or an informal tone.

4 **Choice C** Words in context

A lake whose chemistry barely shifts, even in a drought, is stable. Volatile means the reverse, and porous and scarce do not describe chemical balance.

5 **Choice B** Words in context

Compliments buried under qualifications leave the verdict unclear, which is what equivocal means. Rapturous would mean unreserved praise, and unflinching suggests a bold, direct judgement.

6 **Choice D** Text structure and purpose

The underlined sentence rejects the idea that the flourishes were mere decoration and the next sentence says what they show about mapmakers' attitudes. It does not concede that they were ornament, offer a counterexample, or discuss survey methods.

7 **Choice C** Text structure and purpose

The quick packing and the idea that the room feels like an accusation show Ansel's unease at how little there is to carry. The text gives no reason for the move, no change of mind and no suggestion that anyone else values his things.

8 **Choice A** Text structure and purpose

The first stanza recalls the father's confident knowledge, the second admits the speaker sees no pattern now, and the closing lines note an instinct that persists. No argument, narrative moral or comparison of places appears.

9 **Choice D** Cross-text connections

Text 2 finds that home use between generations, not economic incentive, best predicts survival, so the sociolinguist would say Text 1 underrates the family. The sociolinguist does not rule out schooling or claim that small communities cannot succeed.

10 **Choice B** Central ideas and details

The text rejects the older idea that pigment simply wore away and proposes that switching it off saves energy and so is an advantage. The genes remain intact, and nothing compares cave fish with other fish as study subjects or calls pigment the costliest process.

11 **Choice A** Central ideas and details

The text argues that circulation counts only books leaving and misses the other uses of the building. It does not recommend ending lending, says nothing about most visitors reading newspapers, and says the library may be busier, not quieter.

12 **Choice C** Command of evidence: quantitative

The graph shows the count at 452 in 2021 and 610 in 2023. The rise is 158, which is about 35 percent of 452, so the sharp rise is correct. The other choices misstate the size of the change, its direction, or the years.

13 **Choice D** Command of evidence: quantitative

The claim needs a rise among nearby students and no rise among those farther away. Only the first choice gives both. One choice shows both groups rising, one reverses the direction, and the last compares the groups at a single point rather than over time.

14 **Choice A** Command of evidence: textual

The claim is about a contrast between early and late work, and only the quotation about abandoning the domestic scale states that contrast directly. The others describe the early reputation or a detail of one late scene without contrasting the two.

15 **Choice C** Command of evidence: textual

The claim concerns flying without rest for over a week, and a recorded eleven days with no landing supports it directly. Distance, weight change and shared routes do not show that the bird never stopped.

Answer explanations

SAT Practice Test 1, Reading and Writing, Module 2

Questions 18 to 33

18 Choice B Inferences

If wider roads simply attract more traffic, adding capacity will not cure congestion, and giving people other ways to travel is the logical remedy. The text does not recommend removing roads, and wider roads increased driving rather than lowering it.

19 Choice A Inferences

Aubert does not decide an interpretation in advance and her performances differ widely, so she treats each as something that forms in the moment. The text does not mention imitation, fixed rehearsal or reviews being valued over performance.

20 Choice D Form, structure, and sense

The subject is the plural ceramics, which needs a plural verb. The passive construction are distinguished by completes the sentence. Is distinguished does not agree, and the other options fail to give a complete verb.

21 Choice D Boundaries

With neither or nor, the verb agrees with the nearer subject, engineers, which is plural. Was, is and has been are all singular.

22 Choice A Form, structure, and sense

The participle discovering attaches to the main clause and completes it correctly. Discovered would create a second main verb without a conjunction, she discovered would be a run-on, and and discovering breaks the parallel construction.

23 Choice B Form, structure, and sense

With two routes, the comparative less than is correct, and cost is an amount, not a count. The other options are not idiomatic.

24 Choice C Boundaries

The subject is the singular scientist; along with her colleagues is a parenthetical phrase that does not make the subject plural. The past tense presented agrees with the singular subject and the past event in March. Were presenting and have presented do not fit.

25 Choice D Boundaries

The relative pronoun must show possession of grandmother, so whose is required. Who's means who is, and whom and which do not show possession.

26 Choice A Form, structure, and sense

After had, the past participle risen is needed, and rise is intransitive so raised is wrong. Rose is the simple past, which does not follow had, and rised is not a word.

27 Choice C Transitions

The sentence draws a conclusion from the unevenness just described, so Consequently is the logical link. Instead and Likewise do not fit the relationship, and Previously refers to time.

28 Choice B Transitions

The later critics reached the opposite judgement from the early reviewers, so by contrast is the correct transition. The others suggest an example, a restatement or a consequence.

29 Choice A Transitions

The sentence sets the neglected lute against the much-studied violin, so By contrast is the logical transition.

30 Choice B Rhetorical synthesis

Scale is shown by figures for area and cost, which the first choice gives. The other choices describe the mill's history or the timing of the project without its size.

31 Choice D Rhetorical synthesis

To introduce readers to the author, the choice must name Wen Zhao and say who they are, which only the first does. The others describe the book without identifying its author.

32 Choice C Rhetorical synthesis

The goal is to explain a benefit, and only the first choice links the effect on nitrogen to the reduced need for fertiliser. The others define the practice or give half the explanation.

33 Choice A Transitions

The drought contradicts what the settlers expected, so Instead is the correct transition. Moreover and In particular add or specify, and Therefore suggests a result of the expectation.

Answer explanations

SAT Practice Test 1, Math, Module 1

Questions 1 to 14

1 **Choice B** Data analysis

Club 4 has 40 members and Club 3 has 18, and 40 minus 18 is 22.

2 **Choice B** Percentages

35% of 240 is 0.35 times 240, which is 84.

3 **Choice C** Linear equations

Subtracting 3x from both sides gives 2x − 12 = 10. Adding 12 gives 2x = 22, so x = 11.

4 **Choice A** Linear equations in context

The fee is paid once, so it is the constant 30. The $18 repeats each month, so it multiplies m. The total is 18m + 30.

5 **Choice C** Systems of equations

Adding the equations gives 2x = 20, so x = 10. The value 4 is y.

6 **Choice D** Functions

Each step of 1 in x adds 3 to f(x), so the slope is 3. Then f(1) = 7 gives 3 + 4 = 7, so the intercept is 4.

7 **Choice B** Nonlinear expressions

Multiply the coefficients, 2 times 5 is 10, and add the exponents, 3 plus 4 is 7.

8 **Choice A** Ratios and rates

10 cups of sugar is 5 times 2 cups, so the flour is 5 times 3 cups, which is 15.

9 **Choice C** Triangles

The angles of a triangle add to 180°, so x + 2x + 3x = 6x = 180 and x = 30.

10 **Choice B** Circles

Circumference $2\pi r = 12\pi$ gives r = 6. The area is $\pi r^2 = 36\pi$. The value 12π is the circumference itself.

11 **Choice A** Quadratics

The equation factors as (x − 7)(x + 2) = 0, so x = 7 or x = −2.

12 **Choice D** Probability

40 students take the bus and 20 of them are in Grade 10, so the probability is 20/40 = 1/2. The value 2/5 uses all 50 Grade 10 students.

13 **Choice B** Statistics

The first five scores total 400, and adding 92 gives 492. Divided by 6 the mean is 82.

14 **Choice C** Scatterplots

Substitute x = 7: 6(7) + 40 = 42 + 40 = 82.

Answer explanations

SAT Practice Test 1, Math, Module 1

Questions 15 to 27

15 **Choice A** Inequalities

The total is the 45 already signed up plus 15 per week, and it must be at least 120, so the inequality is 45 + 15w ≥ 120.

16 **Choice D** Functions

f(3) = 2(9) − 9 + 1 = 18 − 9 + 1 = 10.

17 **Choice B** Exponential functions

Doubling every 3 hours means the number of doublings is t/3, so the population is 500 times 2 to the power t/3.

18 **Choice B** Systems of equations in context

Let a be adult tickets. Then 12a + 7(120 − a) = 1,090, so 5a + 840 = 1,090 and a = 50.

19 **Choice C** Radicals

Squaring both sides gives x + 7 = 25, so x = 18. The value 25 is the square of 5, not x.

20 **Choice A** Linear functions

The slope is (9 − 3)/(4 − 1) = 2. Using (1, 3), 3 = 2(1) + b gives b = 1.

21 **28** Linear equations

Let the number be n. Then (3/4)n = 21, so n = 21 times 4/3 = 28.

22 **63** Area

The area of a triangle is half the base times the height: (1/2)(14)(9) = 63.

23 **5** Linear equations

Expanding gives 2x − 6 = 3x − 11. Subtracting 2x and adding 11 gives 5 = x.

24 **30** Percent change

The increase is 12 dollars, and 12 divided by the original 40 is 0.30, which is 30 percent.

25 **8** Quadratics

The equation factors as (x − 8)(x + 2) = 0, so the solutions are 8 and −2. The positive one is 8.

26 **90** Volume

The volume is $\pi r^2 h = \pi(9)(10) = 90\pi$, so k = 90.

27 **12** Right triangles

By the Pythagorean theorem, the other leg is √(169 − 25) = √144 = 12.

Answer explanations

SAT Practice Test 1, Math, Module 2

Questions 1 to 14

1 **Choice D** Functions

f(4) = 3(16) − 32 + 4 = 48 − 32 + 4 = 20.

2 **Choice A** Systems of equations

Adding the equations gives 2x = 20, so x = 10.

3 **Choice C** Quadratics

The equation factors as (x − 6)(x + 8) = 0, so x = 6 or x = −8.

4 **Choice B** Percentages

20% of 160 is 0.2 times 160, which is 32.

5 **Choice D** Exponential functions

Doubling every 6 hours means the number of doublings is t/6, so the population is 300 times 2 to the power t/6.

6 **Choice C** Linear functions

The slope is (16 − 7)/(4 − 1) = 3. Using (1, 7), 7 = 3(1) + b gives b = 4.

7 **Choice A** Scatterplots

Substitute x = 8: 4(8) + 33 = 32 + 33 = 65.

8 **Choice B** Circles in the plane

The center of circle A is (4, −1). Shifting it 4 units left and 3 units down moves the center to (0, −4), and the radius is unchanged, so the equation is x2 + (y + 4)2 = 25.

9 **Choice A** Quadratics

Completing the square: $x^2 - 10x + 22 = (x - 5)^2 - 3$. The smallest value of the square is 0, so the minimum of f is −3.

10 **Choice C** Functions

g(3) = 13, so f(g(3)) = f(13) = 53. Also f(3) = 13, so g(f(3)) = g(13) = 173. The difference is 53 − 173 = −120.

11 **Choice D** Systems of equations

A system has no solution when the lines are parallel but not identical. The second equation is 4 times the left side of the first with a different constant, so a must equal 16/4 = 4.

12 **Choice B** Surface area

Each cube has 6 faces, and gluing two faces together hides 2 of the 12, leaving 10 faces of area s^2. So $10s^2 = 640$, $s^2 = 64$, and s = 8.

13 **Choice A** Circles

Arc length is the radius times the central angle in radians: 12 × π/4 = 3π.

14 **Choice C** Trigonometry

A right triangle with opposite side 5 and hypotenuse 13 has adjacent side 12, so tan A = opposite/adjacent = 5/12.

Answer explanations

SAT Practice Test 1, Math, Module 2

Questions 15 to 27

15 **Choice D** Percentages

Start at 100. After the 20% rise the price is 120, and 30% off 120 leaves 84. That is 16% lower than the original.

16 **Choice B** Absolute value

The inequality means x is within 5 of 8, so $3 \le x \le 13$. That range has 11 integers, counting both ends.

17 **Choice D** Exponential equations

Write 4 as 2^2, so the right side is 2^(2x − 6). Equal bases give x + 1 = 2x − 6, so x = 7.

18 **Choice B** Percentages

70 is 50 more than 20, and 50 / 20 = 2.5, so p = 250.

19 **Choice C** Exponential functions

A year is 12 months, so the value is multiplied by 0.7^(12/6) = 0.7^2 = 0.49 each year. That is a decrease of 1 − 0.49, or 51%.

20 **Choice A** Statistics

Multiplying by 2 multiplies the mean and the standard deviation by 2; adding 8 moves the mean by 8 but does not change the spread. The mean of A is 6.8, so the mean of B is 21.6, which is 14.8 greater. The standard deviation of B is 2 times that of A.

21 **22** Systems of equations

Adding x + y = 29 and x − y = 15 gives 2x = 44, so the larger number is 22.

22 **4** Quadratics

For $ax^2 + bx + c = 0$ the solutions add to $-b/a = -(-12)/3 = 4$.

23 **40** Similar triangles

The scale factor is DE / AB = 6 / 3 = 2. The perimeter of ABC is 20, so the perimeter of DEF is 2 × 20 = 40.

24 **4** Circles in the plane

Complete the square: $(x + 6)^2 + (y + 5)^2 = -45 + 36 + 25 = 16$. The radius is $\sqrt{16} = 4$.

25 **26** Quadratics

The maximum is at $t = -b/(2a) = 8/2 = 4$. Then $h(4) = -1(16) + 32 + 10 = 26$.

26 **10** Quadratics

Exactly one real solution means the discriminant is zero: $b^2 - 4(25) = 0$, so $b^2 = 100$ and b = 10 (taking the positive value).

27 **8** Systems with nonlinear equations

Setting the expressions equal gives $x^2 - 2x + (9 - k) = 0$. One solution means the discriminant is 0: $4 - 4(9 - k) = 0$, so k = 8.